# Spider-Eyes / Multi-Eye (SE/ME) Deterministic Mathematical Specification
# for Efficient RNA 3D Candidate Generation under Compute Constraints

ISO/GxP-Oriented Specification Template

(Author: ____________) (Approver: ____________)

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## Document Control

| Field | Value |
|---|---|
| Document ID | SE-ME-RNA-SPEC |
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| Change Control | All changes shall be recorded in the Revision History table. |

### Revision History

| Version | Date | Summary of Change |
|---|---|---|
| 1.0 | January 11, 2026 | Initial deterministic SE/ME spec (RNA-focused), includes MoM objective, toehold rule, cotranscriptional staging, and ISO/GxP-oriented structure. |

## 1 Introduction

### 1.1 Purpose

This document specifies a *mathematically deterministic* architecture called **Spider-Eyes / Multi-Eye (SE/ME)** to generate multiple plausible RNA 3D structure candidates from sequence and optional constraints, under constrained compute environments.

### 1.2 Scope

This specification defines:

- a deterministic multi-hypothesis generator ("Eyes") that produces discrete structural decisions via categorical "Dice";
- a deterministic fusion/selection controller ("Mosaic-of-Mosaics", MoM) that balances agreement and diversity;
- an RNA-focused mapping from discrete decisions to 3D coordinates via constrained geometric construction and bounded refinement.

### 1.3 ISO/GxP-Oriented Statement (Non-Certification)

This specification is written in a format intended to *support* ISO/GxP documentation patterns (traceability, explicit requirements, verification tests, controlled parameters). It does not itself certify compliance; compliance requires implementation validation, controlled execution, and auditable records.

## 2 Math Uses

The SE/ME method uses:

- Linear algebra (vectors, matrices, projections, norms).
- Probability on finite sets (categorical distributions), with deterministic sampling via seeded PRNG.
- Information theory: Jensen–Shannon divergence to measure cross-eye disagreement.
- Graph theory: secondary-structure graphs and conflict-free pairing.
- Constrained geometry / "fitting" viewpoint: global consistency under local constraints, analogous to locally isometric mapping constraints in origami problems.
- Optimization: bounded, fixed-iteration refinement (deterministic).

## 3 Glossary

| **Term** | **Definition** |
| --- | --- |
| Residue / Nucleotide | One RNA base position, indexed $r \in \{1, \ldots, n\}$ with identity in $\{A, C, G, U\}$. |
| Eye | A parameterized hypothesis generator producing one candidate roll (set of decisions) and its induced 3D structure. |
| Dice | A set of categorical distributions, one per decision variable, producing discrete outcomes. |
| Gate | A deterministic mapping from context to logits that shape Dice distributions. |

| | |
|---|---|
| Cluster | A small group of Gates producing a multi-scale ("fractal") feature vector. |
| Mosaic | The concatenation of Cluster outputs for an Eye, used to drive Dice logits. |
| MoM | Mosaic-of-Mosaics; deterministic fusion/selection across Eyes balancing agreement and diversity. |
| Projection | An injective mapping used to encode the same context from distinct "viewpoints." |
| Cotranscriptional stage | A prefix length $t$ ($1 \le t \le n$) representing sequential folding as RNA length increases. |
| Toehold displacement | A local rearrangement rule where a single-stranded region seeds pairing and then exchanges base pairs in a break-one-form-one manner. |

# 4 Process Model

This section follows the simple process model: **Inputs** $\rightarrow$ **Process** $\rightarrow$ **Outputs**.

## 4.1 Inputs

**REQ-001** The system shall accept an RNA target record:

$$\mathcal{T} = (\mathrm{ID}, s, n)$$

where ID is an identifier string, $s = (s_1, \ldots, s_n)$ is a sequence over $\{A, C, G, U\}$, and $n$ is the length.

**REQ-002** The system shall accept optional constraint inputs:

$$\mathcal{C} = (\mathcal{P}, \mathcal{D}, \mathcal{W})$$

where:

- $\mathcal{P}$ is a set of *allowed* base pairs $(i, j)$ with $1 \le i < j \le n$,
- $\mathcal{D}$ is a set of distance constraints of form $(i, j, d_{ij}, \epsilon_{ij})$,
- $\mathcal{W}$ is a set of weights for penalties (defined later).

**REQ-003** The system shall accept a global determinism seed $S_0 \in \{0, \ldots, 2^{64} - 1\}$.

## 4.2 Outputs

**REQ-010** The system shall output **exactly five** 3D coordinate models per target:

$$\mathbf{X}^{(k)} = \big((x_{r,k}, y_{r,k}, z_{r,k})\big)_{r=1}^{n}, \quad k \in \{1, 2, 3, 4, 5\}.$$

**REQ-011** The system shall output deterministic diagnostics sufficient for audit:

$$\mathcal{A} = (\text{seed log}, \text{parameter set}, \text{MoM scores}, \text{validation report}).$$

(Note: in Kaggle-like contexts, diagnostics may be written as separate files; the submission CSV contains only coordinates.)

### 4.3 Process Overview

The process is composed of:

1. Deterministic Preprocessing.
2. Multi-Eye Candidate Generation (Eyes produce rolls and candidate structures).
3. MoM Fusion and Selection (choose 5 outputs).
4. Deterministic 3D Construction + Bounded Refinement.
5. Output Formatting + Validation.

## 5 Determinism Requirements

**REQ-020** Given the same $(\mathcal{T}, \mathcal{C}, S_0)$, the system shall produce identical outputs $\{\mathbf{X}^{(k)}\}_{k=1}^{5}$ bit-for-bit (up to specified floating-point serialization rules).

**REQ-021** All tie-breaks shall be deterministic:

- For $\arg\max$, choose the *smallest index* among tied maxima.
- For sorting, use stable sort with index as the final key.

**REQ-022** If categorical sampling is used, it shall be performed using the PRNG specified in Section 14 with explicitly derived seeds.

## 6 Preprocessing

### 6.1 Base Pair Compatibility

Define a compatibility indicator:

$$\chi(i,j) = \begin{cases} 1 & \text{if } (s_i, s_j) \in \{(A,U),(U,A),(G,C),(C,G),(G,U),(U,G)\}, \\ 0 & \text{otherwise.} \end{cases}$$

Let the allowed pairing set be:

$$\mathcal{P}^\star = \{(i,j) \mid 1 \le i < j \le n,\ \chi(i,j) = 1\} \cap \mathcal{P}$$

where if $\mathcal{P}$ is not provided, treat $\mathcal{P} = \{(i,j) \mid 1 \le i < j \le n\}$.

### 6.2 Cotranscriptional Staging (Pathway Model)

Define prefix lengths:

$$t \in \{1, 2, \ldots, n\}.$$

At stage $t$, the available indices are $\{1, \ldots, t\}$ and allowed pairs are:

$$\mathcal{P}_t^\star = \{(i,j) \in \mathcal{P}^\star \mid j \le t\}.$$

This formalizes a sequential folding viewpoint; it is motivated by empirical observations that folding can be pathway-dependent and differ from equilibrium refolding.

# 7 Spider-Eyes Projection Encoding

## 7.1 Projection Family

Let $d$ be the context dimension. Define $m$ projections:

$$\mathbb{E} = \{P_1, P_2, \ldots, P_m\}, \quad P_e : \mathbb{R}^d \to \mathbb{R}^d.$$

**REQ-030** Each $P_e$ shall be injective. In finite dimensions, it is sufficient that $P_e$ is invertible.

## 7.2 Deterministic Construction of Projections

Each Eye $e$ is assigned a projection matrix $M_e \in \mathbb{R}^{d\times d}$:

$$P_e(\mathbf{c}) = M_e\mathbf{c}.$$

**REQ-031** $M_e$ shall be full-rank and deterministically generated from $(\mathrm{ID}, S_0, e)$. A valid construction:

- Generate $d^2$ PRNG values in $(0,1)$, map to $(-1,1)$.
- Fill a matrix $A_e$ row-major.
- Set $M_e = A_e + \lambda I$ with $\lambda$ chosen as the smallest nonnegative value such that $\det(M_e) \neq 0$ using step $\Delta\lambda = 10^{-3}$ and max steps $10^6$ (deterministic).

# 8 Multi-Eye GD-FMM Objects (Deterministic)

This section formalizes the objects you described: Dice, Gate, Cluster, Eye, MoM.

## 8.1 Context Tensor

Define a context vector $\mathbf{c}_t \in \mathbb{R}^d$ at stage $t$. A deterministic example:

$$\mathbf{c}_t = \big[\mathrm{onehot}(s_t)\ ;\ t/n\ ;\ \text{local-pair-density}(t)\ ;\ 1\big]$$

zero-padded or linearly projected into $\mathbb{R}^d$ by a fixed matrix.

## 8.2 Gate

Each Gate $g$ maps context to an "angle" (latent) vector:

$$g(\mathbf{u}) = \tanh(A\mathbf{u} + \mathbf{b})$$

where $A \in \mathbb{R}^{h\times d}$ and $\mathbf{b} \in \mathbb{R}^h$ are Eye-specific and deterministically derived from the Eye seed.

## 8.3 Cluster (Multi-Scale / "Fractal" Vector)

A Cluster aggregates $L$ scales using a contraction matrix $R$ with spectral radius $< 1$. Define:

$$\mathrm{Cluster}(\mathbf{c}) = \phi\left(\sum_{\ell=0}^{L-1} R^\ell\, g(\mathbf{c})\right)$$

where $\phi(\cdot)$ is elementwise tanh, and $R$ is deterministically generated and then scaled:

$$R \leftarrow \frac{R}{1 + \|R\|_2}$$

to enforce contraction deterministically.

### 8.4 Eye Mosaic

Eye $e$ has $J$ clusters. Its mosaic is:

$$\mathbf{m}_t^{(e)} = \Big[\text{Cluster}_1^{(e)}(P_e(\mathbf{c}_t)) \ ; \ \ldots \ ; \ \text{Cluster}_J^{(e)}(P_e(\mathbf{c}_t))\Big] \in \mathbb{R}^q.$$

### 8.5 Dice Distributions

Let there be $N$ discrete decision variables (Dice), indexed $i = 1, \ldots, N$. Die $i$ has $K_i$ categories. Define logits:

$$\boldsymbol{\ell}_{i,t}^{(e)} = W_i^{(e)}\mathbf{m}_t^{(e)} + \mathbf{a}_i^{(e)} \in \mathbb{R}^{K_i}.$$

Define categorical probabilities with temperature $\tau_e > 0$:

$$\mathbf{p}_{i,t}^{(e)} = \text{softmax}\left(\boldsymbol{\ell}_{i,t}^{(e)}/\tau_e\right).$$

### 8.6 Deterministic Roll of Dice

**REQ-040** Each Eye shall produce a decision outcome $z_{i,t}^{(e)} \in \{1, \ldots, K_i\}$ deterministically using one of:

- **Mode roll (no PRNG):** $z_{i,t}^{(e)} = \arg\max_k \mathbf{p}_{i,t}^{(e)}[k]$ (smallest-index tie-break).
- **Seeded categorical sampling (deterministic via PRNG):** sample $u \sim U(0, 1)$ from PRNG and choose the first category whose cumulative probability exceeds $u$.

This spec selects **seeded categorical sampling** to preserve controlled diversity while remaining deterministic (Section 14).

## 9 RNA-Specific Decision Encoding

### 9.1 Decision Variables

This spec defines Dice decisions to represent *secondary-structure pairing proposals* at each stage $t$.

For each residue $r \leq t$, define one decision variable:

$$i \equiv r, \quad K_i = t - r + 1,$$

representing "partner index":

$$z_{r,t}^{(e)} \in \{0, r+1, r+2, \ldots, t\}.$$

Interpretation:

- $z_{r,t}^{(e)} = 0$ means residue $r$ is unpaired at stage $t$.
- $z_{r,t}^{(e)} = j$ means propose pairing $(r, j)$.

## 9.2 Deterministic Pairing Repair (Conflict-Free Projection)

Raw proposals may conflict (two residues propose different partners, multiple pairing). Define a deterministic repair operator $\Pi$ that outputs a valid pairing set $S_t^{(e)}$.

Construct candidate pairs:

$$\mathcal{Q}_t^{(e)} = \{(r, z_{r,t}^{(e)}) \mid z_{r,t}^{(e)} \neq 0,\ (r, z_{r,t}^{(e)}) \in \mathcal{P}_t^\star\}.$$

Score each proposed pair by:

$$\mathrm{score}(r,j) = \mathbf{p}_{r,t}^{(e)}[\text{category selecting } j].$$

Sort $\mathcal{Q}_t^{(e)}$ by descending score, tie-break by smaller $(j-r)$ then smaller $r$ then smaller $j$.

Initialize $S_t^{(e)} = \emptyset$ and a matched set $M = \emptyset$. Iterate proposals in sorted order; add $(r,j)$ to $S_t^{(e)}$ iff $r \notin M$ and $j \notin M$. When added, update $M \leftarrow M \cup \{r,j\}$.

**REQ-050** $\Pi$ shall be deterministic and shall output a conflict-free pairing set.

# 10 Toehold-Mediated Strand Displacement Operator (Local Rearrangement)

This spec includes a deterministic local rearrangement rule, motivated by documented toehold-mediated strand displacement with break-one-form-one exchange.

## 10.1 State Representation

A stage-$t$ structural state is represented as a pairing set $S_t$.

## 10.2 Toehold Seed Condition

Let $(a,b)$ be an unpaired candidate pair with $b$ near the 3' end (toehold region) and $a$ in a loop/-exposed region. Define:

$$\mathrm{toehold\_eligible}_t(a,b) = 1$$

iff:

- $a < b \leq t$ and $(a,b) \in \mathcal{P}_t^\star$,
- $b$ is in the last $\kappa$ residues: $b \in \{t-\kappa+1, \ldots, t\}$,
- $a$ is currently unpaired in $S_t$ (i.e., $a \notin M(S_t)$),
- adding $(a,b)$ would create a path of competition with existing helix pairs that can be displaced (defined below).

## 10.3 Break-One-Form-One Displacement Path

Define a displacement path as an ordered sequence of operations:

$$\big((\text{break } (i_1,j_1),\ \text{form } (u_1,v_1)), \ldots, (\text{break } (i_L,j_L),\ \text{form } (u_L,v_L))\big)$$

such that each step breaks exactly one existing pair and forms exactly one new pair.

**REQ-060** A displacement path is valid iff:

- All forms are in $\mathcal{P}_t^\star$.
- After each step, the pairing set remains conflict-free.
- The first form is the toehold seed pair $(a, b)$.
- Each subsequent form shares an endpoint with a previously formed pair, creating a contiguous "invasion" process.

### 10.4 Deterministic Selection of a Displacement Path

If multiple valid paths exist, choose the path minimizing:

$$\sum_{\ell=1}^{L}(v_\ell - u_\ell) \quad \text{then } L \quad \text{then lexicographic order of formed pairs.}$$

This ensures determinism.

## 11 3D Construction from Pairing Set

### 11.1 Coarse-Grained Coordinate Model

Represent each residue $r$ by a single point $\mathbf{x}_r \in \mathbb{R}^3$.

### 11.2 A-Form Helix Idealization (Algorithm Constants)

This spec defines *algorithmic* constants (not a claim of physical exactness):

$$\text{rise } h = 3.0, \quad \text{twist } \theta = 0.6 \text{ rad}, \quad \text{radius } \rho = 10.0.$$

These constants are used to deterministically place paired stems as helices.

### 11.3 Stem Extraction

From pairing set $S_t$, extract stems as maximal runs of consecutive base pairs:

$$(i, j), (i + 1, j - 1), \ldots, (i + \ell, j - \ell).$$

### 11.4 Deterministic Placement Procedure

For each stem, choose a local coordinate frame and place residues along a helix axis:

$$\mathbf{x}_{i+k} = \mathbf{o} + kh\mathbf{u} + \rho\left(\cos(k\theta)\mathbf{v} + \sin(k\theta)\mathbf{w}\right),$$

where $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ is an orthonormal basis and $\mathbf{o}$ is an origin. The basis and origin are deterministically derived from $(\text{ID}, S_0, e, k, \text{stem index})$ by PRNG.

Unpaired residues and loop regions are filled by deterministic interpolation between neighboring placed residues using piecewise-linear segments, then smoothed by fixed-iteration Laplacian smoothing.

### 11.5 Bounded Refinement (Fixed Iterations)

Define an energy:

$$E(\mathbf{X}) = \lambda_1 E_{\text{link}} + \lambda_2 E_{\text{pair}} + \lambda_3 E_{\text{rep}} + \lambda_4 E_{\text{smooth}}.$$

Where:

$$\begin{aligned}
E_{\text{link}} &= \sum_{r=1}^{n-1} \left(\|\mathbf{x}_{r+1} - \mathbf{x}_r\| - d_{\text{link}}\right)^2, \\
E_{\text{pair}} &= \sum_{(i,j)\in S_n} \left(\|\mathbf{x}_j - \mathbf{x}_i\| - d_{\text{pair}}\right)^2, \\
E_{\text{rep}} &= \sum_{1\le i<j\le n} \max\{0, d_{\min} - \|\mathbf{x}_j - \mathbf{x}_i\|\}^2, \\
E_{\text{smooth}} &= \sum_{r=2}^{n-1} \left\|\mathbf{x}_r - \frac{\mathbf{x}_{r-1} + \mathbf{x}_{r+1}}{2}\right\|^2.
\end{aligned}$$

Algorithmic constants:

$$d_{\text{link}} = 6.0, \quad d_{\text{pair}} = 15.0, \quad d_{\min} = 3.0.$$

Run gradient descent for exactly $T = 50$ iterations with step $\eta = 10^{-3}$ using deterministic finite differences:

$$\frac{\partial E}{\partial x_{r,d}} \approx \frac{E(\mathbf{X} + \epsilon \mathbf{e}_{r,d}) - E(\mathbf{X} - \epsilon \mathbf{e}_{r,d})}{2\epsilon}, \quad \epsilon = 10^{-4}.$$

**REQ-070** Refinement shall use fixed $(T, \eta, \epsilon)$ to preserve determinism.

## 12 MoM: Fusion, Coherence, Diversity, Selection

### 12.1 Per-Eye Distributions

Each Eye produces $\mathbf{p}_{i,n}^{(e)}$ for each decision variable $i$ at full length $n$.

### 12.2 Disagreement (Cross-Eye Jensen–Shannon)

For each die $i$, define the average JS divergence across eyes:

$$D_i = \frac{2}{E(E-1)} \sum_{e<e'} \text{JS}\left(\mathbf{p}_{i,n}^{(e)}, \mathbf{p}_{i,n}^{(e')}\right),$$

with:

$$\text{JS}(p, q) = \frac{1}{2}\text{KL}(p\|m) + \frac{1}{2}\text{KL}(q\|m), \quad m = \frac{p+q}{2}.$$

Define global disagreement:

$$\text{Disagreement} = \frac{1}{N}\sum_{i=1}^{N} D_i.$$

### 12.3 Novelty (Behavioral Diversity)

Let each Eye have a behavior vector:

$$\mathbf{b}^{(e)} = \frac{1}{n}\sum_{t=1}^{n} \mathbf{m}_t^{(e)}.$$

Define novelty as average pairwise cosine distance:

$$\text{Novelty} = \frac{2}{E(E-1)}\sum_{e<e'}\left(1 - \frac{\langle \mathbf{b}^{(e)}, \mathbf{b}^{(e')}\rangle}{\|\mathbf{b}^{(e)}\|\,\|\mathbf{b}^{(e')}\|}\right).$$

### 12.4 Task Score (RNA Plausibility)

Define a per-Eye task score:

$$\text{TaskScore}^{(e)} = -E(\mathbf{X}^{(e)})$$

where $\mathbf{X}^{(e)}$ is the Eye's constructed/refined 3D model from $S_n^{(e)}$.

### 12.5 MoM Objective and Eye Weights

Define the MoM objective:

$$\mathcal{E} = \alpha \cdot \text{Disagreement} - \beta \cdot \text{Novelty} + \gamma \cdot \frac{1}{E}\sum_{e=1}^{E} \text{TaskScore}^{(e)}.$$

with fixed constants $\alpha, \beta, \gamma > 0$.

Define Eye weights:

$$w_e = \frac{\exp(\delta \cdot \text{TaskScore}^{(e)})}{\sum_{e'} \exp(\delta \cdot \text{TaskScore}^{(e')})}$$

with fixed $\delta > 0$.

### 12.6 Selection of Five Outputs

Construct candidate list:

$$\mathcal{K} = \{(e, \mathbf{X}^{(e)}) \mid e = 1, \ldots, E\}.$$

Rank candidates by:

$$\text{RankScore}(e) = w_e \cdot (-E(\mathbf{X}^{(e)})).$$

Select the top 5 candidates. If fewer than 5 eyes exist, deterministically generate additional candidates by re-rolling with the same Eye using seed offsets $+1, +2, \ldots$ until 5 are produced.

**REQ-080** The system shall output exactly 5 models and shall document the selection ordering and scores.

## 13 Dynamic Eye Assignment (Spawn / Merge / Retire)

This section makes your "spawn/merge" idea deterministic.

### 13.1 Limits

**REQ-090** The number of Eyes shall satisfy:

$$E_{\min} = 2 \le E \le E_{\max} = 12.$$

### 13.2 Spawn Rule

Partition Dice indices into $B$ contiguous blocks. Compute block disagreement:

$$\text{Disagreement}(b) = \frac{1}{|I_b|} \sum_{i \in I_b} D_i.$$

Spawn a new Eye if:

$$\exists b : \text{ Disagreement}(b) > \theta_{\text{spawn}} \quad \text{and} \quad E < E_{\max}.$$

The spawned Eye parameters are deterministically copied from the current best Eye and perturbed by fixed offsets derived from PRNG.

### 13.3 Merge / Retire Rule

Compute pairwise cosine distance of behavior vectors. If there exist $e \neq e'$ with:

$$1 - \cos(\mathbf{b}^{(e)}, \mathbf{b}^{(e')}) < \theta_{\text{merge}},$$

then deterministically retire the one with lower RankScore, provided $E > E_{\min}$.

## 14 PRNG Specification

### 14.1 Seed Derivation

Define a deterministic 64-bit hash H of byte strings (FNV-1a 64-bit):

$$\text{H(bytes)} = ((\cdots((\text{offset} \oplus b_1) \cdot p \oplus b_2) \cdot p \cdots) \oplus b_L) \cdot p \bmod 2^{64}$$

with:

$$\text{offset} = 14695981039346656037, \quad p = 1099511628211.$$

Eye seed:

$$S_e = \text{H}(\text{ID} \parallel S_0 \parallel e)$$

where $\parallel$ denotes byte concatenation with a fixed delimiter.

### 14.2 LCG Generator

State update:

$$x_{t+1} = (a x_t + c) \bmod 2^{64}$$

with:

$$a = 6364136223846793005, \quad c = 1442695040888963407.$$

Uniform in $(0, 1)$:

$$u_{t+1} = \frac{x_{t+1}}{2^{64}}.$$

### 14.3 Categorical Sampling

Given probabilities $(p_1, \ldots, p_K)$ and $u \in (0, 1)$:

$$\text{choose smallest } k \text{ such that } \sum_{j=1}^{k} p_j \geq u.$$

## 15 Verification and Validation (Spec-Level)

### 15.1 Verification Requirements

**TEST-001** Determinism Test: rerun twice with same inputs and confirm identical outputs.
**TEST-002** Shape Test: confirm output contains exactly 5 models for each target and all residues.
**TEST-003** Numeric Test: confirm no NaNs/Infs and values within representable float range.
**TEST-004** Pairing Validity Test: confirm $\Pi$ produces conflict-free pairs.
**TEST-005** Audit Log Test: confirm seeds and parameter IDs are recorded.

### 15.2 Traceability Matrix (Minimal)

| Requirement | Verification |
|---|---|
| **REQ-020**,**REQ-021**,**REQ-022** | **TEST-001** |
| **REQ-010**,**REQ-080** | **TEST-002** |
| **REQ-050**,**REQ-060** | **TEST-004** |
| **REQ-011** | **TEST-005** |

## 16 How This Supports RNA 3D Folding Under Compute Limits (Non-Normative)

- **Pathway sensitivity:** cotranscriptional staging ($t = 1 \to n$) models order effects instead of only end-state equilibrium.
- **Local-to-global:** clusters encode local context; MoM enforces global consistency like a "fitting" problem under constraints.
- **Efficient rearrangement modeling:** toehold operator provides a computable local rule for large structural rearrangements without full all-atom simulation.
- **Five outputs naturally:** Multi-Eye generation is explicitly designed to produce five diverse candidates with deterministic selection.

## 17 References (Supplied Documents)

## References


[1] A. M. Yu et al. *Computationally Reconstructing Cotranscriptional RNA Folding Pathways from Experimental Data Reveals Rearrangement of Non-Native Folding Intermediates*. Supplied PDF: `Conscffriptopjnfoldiong.full.pdf`.

[2] (Authors as in supplied PDF). *Ligand-state / ZTP riboswitch cotranscriptional behavior and decision window*. Supplied PDF: `Ligenstate.full.pdf`.

[3] K. E. Watters, E. J. Strobel, A. M. Yu, et al. *Cotranscriptional folding in a riboswitch system and differences from equilibrium refolding*. Supplied PDF: `watters_strobel_yu_cotranscriptional_riboswitch_nsmb_3316.pdf`.

[4] S. I. (as supplied). *Origami mathematics: locally isometric mappings, constraints, and Lagrangian framing*. Supplied PDF: `Ornagomi Math igomsn_january-february2022.pdf`.

[5] User-supplied. *Spider-Eyes / Multi-Eye GD-FMM concept notes*. Supplied DOCX: `Spidere eyes.docx`.

[6] User-supplied. *Toy prototype: multi-eye GD-FMM*. Supplied script: `# multi_eye_gdfmm.py`.


## A Appendix A: Parameters (Fixed Defaults)

| Parameter | Default |
|---|---|
| Number of initial eyes $E_{\min}$ | 2 |
| Maximum eyes $E_{\max}$ | 12 |
| Projection dimension $d$ | Implementation-defined constant |
| Clusters per eye $J$ | 6 to 12 (implementation fixed) |
| Cluster depth $L$ | 4 |
| Helix constants $(h, \theta, \rho)$ | $(3.0, 0.6, 10.0)$ |
| Refinement iterations $T$ | 50 |
| Gradient step $\eta$ | $10^{-3}$ |
| Finite difference $\epsilon$ | $10^{-4}$ |
| Spawn threshold $\theta_{\text{spawn}}$ | 0.25 |
| Merge threshold $\theta_{\text{merge}}$ | 0.05 |
| MoM weights $(\alpha, \beta, \gamma, \delta)$ | Implementation-fixed constants |

## B Appendix B: Chat Text (Best-Effort Verbatim Capture)

**Note:** Screenshots are represented as bracketed placeholders. The intent is to preserve the text content exchanged.

```
USER: I have no idea how to set up a Kaggle notebook for a submission to the Stanford RNA
     3D Folding Part 2.

USER: (Provided code that writes /kaggle/working/submission.csv and prints shape/head.)
```

```
USER: no find and replace only full file replacement.
      No Multi-step/multi item instructions only 1 Item, its associated Steps, with a
          validation check.
      No assuming User Intentions.
      Always explicitly recommend an option when presenting options.
      Always recommend next steps.
      The User Is Not a Developer... Never assume...

USER: I tried that and stopped it after an hour.

USER: [Screenshot: Kaggle notebook output shows wrote submission.csv but confusion about
    Output panel.]

USER: I restarted from create new notebook... I do not see "Add Data".

USER: OK from now on let's do this literally one step at a time with verifications in
    between...

USER: [Screenshot: Add Input dialog]

USER: I did this - then what.

USER: Verified: I can see sample_submission.csv.

USER: OK... so there you go?

USER: I made sure the internet was off then clicked save.

USER: Is this what you're looking for?

USER: I see submission.csv in Output.

USER: Submission running for ~22 minutes... (then) finished and got score 0.058.

USER: OK so how do I upload my own .csv for submission

USER: I got a Submission Scoring / Format Error.

USER: Here is the code from the notebook that copies uploaded CSV from /kaggle/input/...
    to /kaggle/working/submission.csv.

USER: (Asks if they should click Submit to competition)

USER: I was thinking ... Spider eyes... Multi-Perspective Projection Encoding (Spider-
    Eyes)
      E = {P1,...,Pm}
      Ei(Gk)=Pi(Gk)
      Properties: Injective, Class-Preserving, Similarity-Distorting...

USER: Multi-Eye GD-FMM: concept
      Objects: Dice, Gate, Cluster, Eye, Mosaic-of-Mosaics (MoM)
      Energy & consensus: coherence via JS divergence; objective E = alpha*Disagreement -
          beta*Novelty + gamma*TaskScore
```

```
     Dynamic assignment: spawn/merge/retire...

USER: I accept your recommendations but I need you to turn what you've learned and what
    you think about spider eyes into a complete and full mathematically deterministic
    specification...
     ...support RNA 3D Folding...

USER: Uhm can you turn all of the into a latex

USER: Blockers about compute, dataset size, 5 models, residue numbering, coordinate
    clipping...
     Thought experiment: localized cluster behavior, "Lincoln logs", Higgs-field analogy,
          IRF/FMM/Spider Eyes, UF-Core layers...

USER: Now consider if there is anything meaningful in these documents... to improve
    accuracy/predictability/compute.

USER: I want you to take what we have just learned and turn it into a New Version Spider-
    Eyes / Multi-Eye LaTeX spec
     fully mathematically deterministic, ISO/GXP compliance,
     math uses section, glossary, introduction,
     Inputs-Process-Outputs,
     references to supplied docs,
     and all chat text in an appendix.
```